

Remote Bone Health Service for Osteoporosis Screening in High-Risk Men A Cluster Randomized Clinical Trial

Cathleen Colón-Emeric, MD, MHS; Richard Lee, MD, MPH; Kenneth W. Lyles, MD; Leah L. Zullig, PhD;
Richard Sloane, MS; Carl F. Pieper, DrPH; Richard E. Nelson, PhD; Robert A. Adler, MD

IMPORTANCE Evidence supporting osteoporosis screening in men is limited. Efficient models that promote osteoporosis screening and adherence in primary care are needed.

OBJECTIVE To test the impact of a centralized, remote bone health service (BHS) on screening, treatment, adherence, and bone density in older men with fracture risk factors.

DESIGN, SETTING, AND PARTICIPANTS This cluster randomized clinical trial involved 39 primary care teams in 2 Veterans Affairs (VA) Health Systems that included 3112 male veterans aged 65 to 85 years who had at least 1 fracture risk factor but had no prior fractures. Data were analyzed from July through November 2025.

INTERVENTION Eligible men in medical teams randomized to BHS were invited to undergo dual-energy x-ray absorptiometry (DXA) scan, followed by an electronic consult to their primary care clinician with additional recommendations. A nurse care manager entered orders, obtained test results, provided patient education, and monitored adherence over the phone. Primary care teams randomized to usual care received osteoporosis education and VA practice guidelines.

MAIN OUTCOMES AND MEASURES DXA screening rates, osteoporosis treatment, persistence, and adherence were compared between patients in BHS and usual care. A random subset of patients in each team had a DXA 24 months after team enrollment, regardless of whether they had been screened during the intervention.

RESULTS Of the 3112 randomized participants (mean [SD] age, 73.3 [5.3] years; 1260 [40.4%] were Black; and 1748 [56%] were White), 49.2% (830 of 1688) in the BHS group were screened vs 2.3% (33 of 1424) in the usual care group ($P < .001$). More than half of those screened (441 [51.1%]) had osteopenia or osteoporosis. One hundred fifty-seven patients (84.4%) in the BHS group initiated osteoporosis treatment and achieved high levels of adherence with a mean of 91.7% of subsequent days covered and a high persistence with a mean 657 (SD, 366) days over 2 years of follow-up. The mean femoral neck T-score 2 years after team initiation in a random subset favored BHS vs usual care (-0.55 vs -0.70 , $P = .04$).

CONCLUSIONS AND RELEVANCE This cluster randomized clinical trial found that the BHS model was associated with significantly improved osteoporosis screening, treatment, and adherence compared with usual care, with high patient and clinician acceptance. Selecting men for osteoporosis screening based on clinical risk factors before a fracture has occurred has a high screening yield. This approach requires validation in other clinical settings and with longer follow-up to determine impact on fractures.

TRIAL REGISTRATION ClinicalTrials.gov Identifier: [NCT04079868](https://clinicaltrials.gov/ct2/show/study/NCT04079868)

JAMA Intern Med. 2025;185(10):1218-1224. doi:10.1001/jamainternmed.2025.4150
Published online August 25, 2025.

[← Invited Commentary page 1225](#)

[+ Supplemental content](#)

Author Affiliations: Geriatrics Research Education and Clinical Center and Center of Innovation to Accelerate Discovery and Practice Transformation, Durham VA Health System, Durham, North Carolina (Colón-Emeric, Lee, Lyles, Zullig, Sloane); Duke University School of Medicine, Durham, North Carolina (Colón-Emeric, Lee, Lyles, Zullig, Pieper); Informatics, Decision Enhancement, and Analytic Sciences Center, VA Salt Lake City Healthcare System, Salt Lake City, Utah (Nelson); Division of Epidemiology, University of Utah, Salt Lake City (Nelson); Central Virginia Veterans Affairs Health Care System and Virginia Commonwealth University, Richmond (Adler).

Corresponding Author: Cathleen Colón-Emeric, MD, MHS, 508 Fulton St, GRECC 182, Durham, NC 27705 (cathleen.colonemeric@duke.edu).

jamainternalmedicine.com

At age 50 years, 1 in 5 men can expect to sustain an osteoporotic fracture in their remaining lifetime with excess mortality rates of 2.7 to 22.3 per 100 person-years, exceeding that of other conditions for which screening is recommended.¹ Osteoporosis screening with a dual energy x-ray absorptiometry (DXA) scan is noninvasive and inexpensive. Safe, effective treatments are available, with first-line medications costing less than \$50 a year. Economic modeling suggests that screening men for osteoporosis is highly cost-effective.²⁻⁵ Despite this strong rationale, screening is rarely considered for men in the US before they have already sustained a fracture, with primary screening rates of less than 10%.⁶⁻⁹

This low rate is partly due to a lack of randomized trials, leading to inconsistent clinical practice guidelines. The US Preventive Services Task Force and the VA National Center for Health Promotion have concluded that current evidence is insufficient to recommend for or against osteoporosis screening in men without prior fractures.¹⁰ Professional society guidelines recommend screening based on the rationale above but vary in how to select men for screening.¹¹⁻¹⁴ For example, the Bone Health and Osteoporosis Foundation guidelines recommend screening all men older than 70 years whereas the VA undersecretary guidelines suggest screening men older than 65 years with 1 or more risk factors for fracture.¹⁵ A retrospective cohort study using VA and Centers for Medicare & Medicaid (CMS) data for more than 5 million male veterans found that primary osteoporosis screening was likely effective in reducing fractures among those with the conditions listed in the VA undersecretary guidelines, with hazard ratios ranging from 0.77 to 0.90.⁶

Even when guidelines are clear, osteoporosis treatment rates remain low. Multiple randomized trials demonstrate that treatment with bisphosphonates after major osteoporotic fracture reduces secondary fractures and mortality in both men and women.¹⁶⁻¹⁸ However, fewer than one-third are treated.¹⁹ Among those who are treated, long-term adherence remains a concern with fewer than half of Medicare beneficiaries achieving levels associated with fracture reduction.²⁰ Interventions to improve screening and adherence including patient education, clinician reminders, and audit and feedback result in little or no improvement in most studies.^{21,22} Fracture liaison services, in which a centralized team coordinates osteoporosis care, have been effective in improving treatment after fracture but have not been widely studied for primary prevention, ie, before the first fracture has occurred.^{23,24}

Thus, there is a need for more evidence on the impact of primary osteoporosis screening on fracture risk in men, which men to select for screening, and efficient care models that improve screening rates and treatment adherence. The objective of this study was to compare the impact of novel screening models vs usual care on patient-level outcomes strongly associated with fracture rates, including proportion screened, proportion treated, medication adherence, and bone density at 2 years.

Methods

The study protocol has been published previously (NCT04079868).²⁵ The statistical analysis plan is available in

Key Points

Question Does a centralized, remote bone health service (BHS) improve osteoporosis screening, treatment, adherence, and bone density among older men with fracture risk factors?

Findings This cluster randomized clinical trial of 39 primary care clinicians and 3112 patients found that a remote BHS significantly improved screening, treatment, and adherence compared with usual care, with a higher average femoral neck bone density at 2 years. The approach of using fracture risk factors to select men for screening yields a comparable proportion of men diagnosed with osteopenia or osteoporosis with screening of women 65 years or older.

Meaning This cluster randomized clinical trial supported selecting men for osteoporosis screening based on fracture risk factors and found that a centralized screening model is associated with improved screening, treatment, and medication adherence rates.

Supplement 1. This was a pragmatic, single-blind (for outcomes assessment), group randomized trial of male veterans aged 65 to 85 years without prior fracture who qualified for osteoporosis screening under VA undersecretary guidelines.¹⁵ Primary care physicians at 2 medical centers were recruited from 39 medical teams; specialty clinics (eg, those serving women or younger Gulf War veterans), part-time clinicians (<80% effort), and trainees were excluded. In the VA, a primary care team consists of a single medical practitioner and support staff caring for a unique panel of patients. Stratified, block randomization at the team level was employed by a statistician unaware of team identity using the RAND function in SAS. Teams were stratified by medical center, hospital vs community-based clinic, and clinician type (physician vs advanced practice physician). All study procedures were considered standard clinical care or minimal risk, and a waiver of patient informed consent was approved by the Durham and Richmond VA Medical Center institutional review boards. The Consolidated Standards of Reporting Trials (CONSORT) guidelines were used. Race was determined from the electronic health record (EHR) demographics page.

Teams were initially randomized into 3 groups: (1) usual care; (2) a practice management model in which teams were provided with additional tools to promote osteoporosis screening, or (3) a bone health service (BHS) model, in which a centralized team used an EHR case-finding tool and a nurse care manager coordinated screening and treatment monitoring. Three months after randomization on December 26, 2019, and before any patient had been screened, the study was halted for 6 months due to the COVID-19 pandemic. Resulting workflow changes in primary care rendered the practice management model impracticable, so teams initially randomized to this arm were rerandomized to usual care or BHS.

Within these teams, male veterans aged 65 to 85 years eligible for primary osteoporosis screening were identified from the EHR using active medical problem list, *International Statistical Classification of Diseases and Related Health Problems, Tenth Revision (ICD-10)* diagnosis codes, and pharmacy records from the prior 2 years. Eligibility criteria included no prior fracture or osteoporosis medication and at least 1 risk

factor listed in the VA undersecretary screening guidelines: hyperthyroidism, hyperparathyroidism, rheumatoid arthritis, alcohol dependence, chronic lung disease, chronic liver disease, stroke, parkinsonism, prostate cancer, smoking, diabetes, pernicious anemia, gastrectomy, or high-risk medication use of 3 or more months in the past 2 years (traditional anti-epileptics, glucocorticoids, androgen deprivation therapy).¹⁵ Patients were excluded if they had end-stage kidney disease or were receiving palliative care. When more than 100 men in a team were eligible, the 100 with the lowest Osteoporosis Self-Assessment Tool (OST) scores were selected for inclusion due to limited BHS nurse capacity. The OST is a risk-stratification tool that has previously been validated among older male veterans.²⁶ This process limited the sample to the highest-risk tertile (32.5% of eligible patients) based on OST variables (older age and lower body weight).

The BHS model included osteoporosis screening and medication adherence promotion. To promote screening, the BHS nurse identified eligible veterans using an EHR dashboard and invited them via letter and up to 3 phone calls. If the veteran agreed, the nurse placed DXA orders and coordinated scheduling. When results were received, an electronic consult was jointly completed by the nurse and an osteoporosis expert. Those meeting treatment thresholds (osteoporosis at any site or T-score between -1 and -2.4) plus elevated 10-year fracture risk measured by the Fracture Risk Assessment Tool ($>20\%$ major fracture, 3% hip fracture) were recommended osteoporosis medications (oral alendronate in absence of a contraindication as per VA formulary guidance), vitamin D, and dietary or supplemental calcium. Once approved by the primary care clinician, the nurse placed orders, provided patient education over the phone, and mailed written instructions. To promote adherence, the nurse called patients at 1, 6, and 12 months to address common barriers. For example, patients reporting difficulty remembering medications received automated text reminders, and those reporting adverse gastrointestinal effects were referred for parenteral zoledronic acid. Teams assigned to usual care were given the undersecretary guidelines and osteoporosis management education.

Patients were followed up for 2 years after their team start dates to allow sufficient time for screening to occur in usual care patients seen annually and for changes in bone mineral density to become evident among patients starting osteoporosis medications. The usual care team start date was defined as the date the clinician received the undersecretary guidelines. The BHS team start date was the date letters inviting patients for DXA screening were mailed. Teams were staggered over 12 months to accommodate DXA and staffing availability (first team start date, September 1, 2020; last follow-up, March 6, 2024). Data were analyzed from July to November 2024.

Outcomes and covariates were extracted from the EHR via a structured query language program or chart abstraction by a study team member blinded to team assignment and confirmed by a second team member. The coprimary outcomes were the proportion of eligible veterans screened (prespecified important difference, 25%), and T-scores at the femoral neck in a random subset of patients from each team at 2 years.

To obtain T-scores, the list of patients eligible at baseline was ordered using a random number generator, and staff called patients until 25 in each team had scheduled a DXA scan or the list was exhausted. Patients were eligible for the outcome DXA scan regardless of whether they had been screened during the study period, so long as the prior DXA was completed at least 18 months previously to abide by CMS guidelines. Medication persistence and adherence were assessed using pharmacy-dispensing data. Persistence was defined as the time from the first fill date through the last fill date plus the number of days supplied in the last fill. Adherence was defined as the proportion of days covered by dispensed medication, calculated as the total number of days of medication prescribed divided by the total duration of use.^{27,28} Non-VA prescriptions were identified by chart abstraction. Fractures identified in the VA EHR using ICD-10 codes and verified by review of radiology records were an exploratory outcome due to limited power and incomplete ascertainment; 37% of emergency visits among VA patients occur in the community.²⁹ Missing data were either not at random (eg, femoral neck T-scores not calculated due to hip arthroplasty in 3% of sample) or occurred in less than 10% (OST, 0.3%, body mass index, 9.6%), and were handled with a missing indicator.

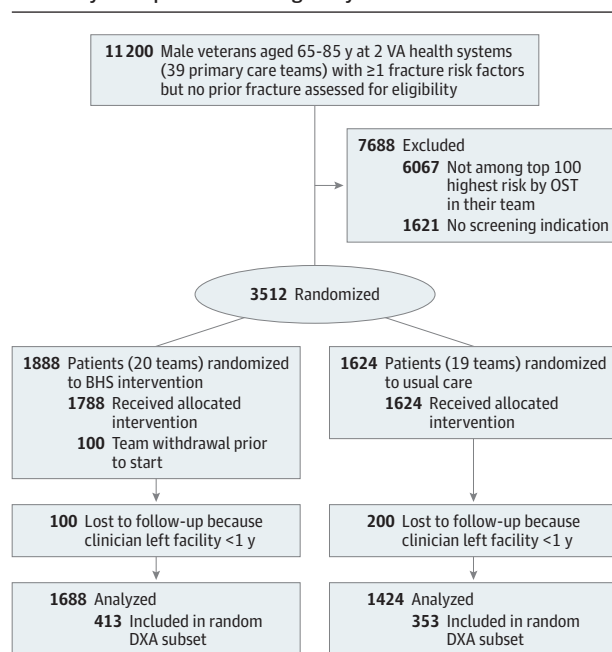
For the primary aim, Poisson regression was used to compare screening rates using the time patients remained in the study as the offset. Patients were censored at death or their last VA encounter date. The unit of analysis was the patient accounting for clustering within teams. Patients remained in the group to which their team was randomized regardless of whether they transitioned to another team. Model covariates included age, race, body mass index, health system, and hospital vs community clinic. The 2-year T-score at the femoral neck was compared using ordinary least squares regression, adjusting for age, race, body mass index, fracture risk factors listed in the VA undersecretary guidelines, time on study, and clustering within teams. To account for multiple statistical testing, the Bonferroni corrected 2-sided *P* value of .025 was set as the level of significance for the coprimary outcomes. Due to low event rates, the difference in proportion with fracture was calculated using an unadjusted χ^2 as an exploratory analysis. SAS version 9.4 was used for all analyses (SAS Institute Inc). Sample sizes of the overall cohort and 2-year DXA subset scan results were calculated for the original 3-arm design to provide at least 0.90 power to detect a clinically meaningful 20% absolute increase in screening rates and 0.10 difference in T-scores between either of the intervention groups and usual care, determined from comparable studies in women³⁰ and with input from operational partners.

Results

Of the 95 eligible primary care clinicians, 39 were randomized to 1 of the 3 original study arms. The most common reason for declining participation was concern about workload if randomized to the practice management arm. Following the 6-month COVID-19 pause, clinicians randomized to this arm were asked to reconsent for randomization to usual care or BHS.

Table 1. Baseline Comparison of Relevant Characteristics of Enrolled Primary Care Teams

Characteristic	No. (%)	
	Usual care (n = 19)	Bone health service (n = 19)
Location		
Urban medical center	9 (48)	9 (48)
Community-based clinic	10 (53)	10 (52)
Clinician		
Doctor of medicine or osteopathic medicine	15 (79)	14 (74)
Advanced practice clinician	4 (21)	5 (16)

Figure. Patient and Team-Level Flow in the Models of Primary Osteoporosis Screening Study

BHS indicates bone health services; DXA, dual energy x-ray absorptiometry; OST, Osteoporosis Self-Assessment Tool; VA, Veteran Affairs.

One clinician originally randomized to BHS withdrew prior to the study restart, so 38 teams were ultimately allocated; 19 to usual care and 19 to BHS. Characteristics of enrolled teams were similar between arms (Table 1).

Within participating teams there were 11 200 male veterans aged 65 to 85 years potentially eligible for primary osteoporosis screening. Of these, 9579 (85.5%) had 1 or more indication for screening according to the VA undersecretary guidelines, and 163 (1.7%) had prior DXA. Nearly one-third (3595 [32.1%]) of those eligible were prescribed a high-risk medication (androgen deprivation therapy, traditional anti-epileptic drug, corticosteroid). For teams with more than 100 eligible patients, the 100 with the lowest OST score were included, resulting in 1788 allocated to the BHS arm and 1624 allocated to usual care (Figure). One clinician in the BHS arm left the facility before their patients had been offered screening, and 2 clinicians in the usual care arm left the facility within 1

Table 2. Baseline Comparison of Relevant Characteristics of Enrolled Patients

Characteristic	No. (%) of patients		P value
	Bone health service (n = 1688)	Usual care (n = 1424)	
Age, mean (SD), y	73.0 (5.2)	73.8 (5.4)	<.01
Race ^a			
Black or African American	695 (41.1)	565 (39.7)	.32
White	930 (55.1)	818 (57.4)	
Other	18 (1.1)	17 (1.2)	
Unknown	45 (2.7)	24 (1.7)	
Rural	541 (32.1)	485 (34.1)	.23
BMI, mean (SD)	26.4 (4.8)	26.4 (4.2)	.86
OST, mean (SD) ^b	1.52 (2.7)	1.44 (3.3)	.43
Corticosteroids	516 (30.6)	344 (24.2)	<.001
Seizures	34 (2.4)	41 (2.4)	1.0
Alcohol use disorder	216 (12.8)	185 (13.0)	.87
Tobacco use	422 (25.0)	376 (26.4)	.38
Diabetes	666 (39.5)	547 (38.4)	.55
Rheumatoid arthritis	29 (2.0)	25 (1.4)	.27
Hyperparathyroidism	14 (0.8)	7 (0.5)	.28
Hyperthyroidism	5 (0.3)	3 (0.2)	.73
Stroke	128 (7.6)	106 (7.4)	.89
Parkinson disease	43 (2.6)	33 (2.3)	.72
Prostate cancer	220 (13.0)	202 (14.2)	.37
Chronic lung disease	399 (23.6)	241 (16.9)	<.001
Chronic liver disease	68 (4.0)	78 (5.5)	.06
Chronic kidney disease stage 4 or 5	2 (0.1)	8 (0.6)	.05
Pernicious anemia or gastrectomy	51 (3.0)	50 (3.5)	.40

Abbreviations: BMI, body mass index, calculated as weight in kilograms divided by height in meters squared; OST, Osteoporosis Self-Assessment Tool.

^a Race was determined by from the electronic health record demographics page. Other included American Indian, Asian, and multiple races.

^b OST is a validated formula that predicts risk of osteoporosis based on current weight and age. For males, OST equals (weight [kg] – age [years])/5. Scores ranging from –20 to –1 are considered high risk, from –1 to 4 are considered moderate risk, and more than 4 are low risk.²⁶

year of their team start date such that their patients did not have sufficient opportunity to have been offered screening, with loss to follow-up of 300 patients (9.6%).

Of the 3112 patients randomized (mean [SD] age, 73 [5.3] years; 1260 [40.4%] were Black; and 1748 [56%] were White), 224 patients (7.2%) died and 162 patients (5.2%) left the facility. Overall, patients were followed up for a mean of 1.5 years. Baseline patient characteristics are in Table 2.

The BHS significantly increased screening among eligible patients compared with usual care (49.2% [830 of 1688] vs 2.3% [33 of 1424]; $P < .001$; Table 3). The proportion of normal DXA results were higher in BHS teams than usual care teams; however, the overall yield of patients meeting the fracture risk threshold for recommending osteoporosis treatment was similar (186 [22.4%] BHS vs 9 [27.2%] usual care). Among patients in the BHS group who met the treatment threshold, 157 (84.4%) initiated osteoporosis medications: 114 (72.4%) alendronate,

Table 3. Bone Health Outcomes in Bone Health Service and Usual Care Groups

Outcomes	No. (%) of patients		P value
	Bone health service (n = 1688)	Usual care (n = 1424)	
Patients screened	830 (49.2)	33 (2.3)	<.001
Screening results			
Normal	414 (49.9)	10 (30.3)	
Osteopenia treatment threshold			
Under	230 (27.7)	14 (42.4)	<.001
Over	119 (14.3)	8 (24.2)	
Osteoporosis	67 (8.1)	1 (3.0)	
Medication initiation, No. treated/No. eligible (%) ^a	157/186 (84.4)		
Duration of treatment, mean (SD), d ^a	657 (366)		
Proportion of days covered with osteoporosis medication, mean (SD), % ^a	0.92 (0.24)		
Fractures	31 (1.8)	29 (2.0)	.69
Patients DXA 2-y measures, No.	413	353	
Femoral neck T-score, mean (SD) ^b	-0.55 (1.28)	-0.70 (1.25)	.04 ^c
Spine T-score, mean (SD) ^b	1.16 (1.95)	1.37 (1.96)	.10 ^c
Total hip T-score, mean (SD) ^b	0.07 (1.3)	-0.05 (1.3)	.13 ^c

^a Unable to compare with the usual care group due to small number of patients meeting treatment threshold (n = 9).

^b Initially eligible patients were recontacted at 2-year in random order until 25 per team were scheduled for dual energy x-ray absorptiometry (DXA; mean T-scores for these individuals). A T-score of negative 2.5 or less indicates osteoporosis; a T-score between -1 and -2.4, osteopenia.

^c P value derived from group effect nested within clinics.

34 (21.6%) zoledronate, 5 (3.4%) denosumab, and 4 (2.6%) anabolic agent, with high levels of persistence (mean, 657 [SD, 366] days) and adherence (mean proportion of days covered, 91.7%). In the usual care group, only 9 screened patients were eligible for treatment, precluding statistical comparisons of treatment and adherence rates. However, the adherence rate observed in BHS patients was substantially higher than the osteoporosis medication adherence rate among all veterans in the 2 participating facilities immediately prior to study initiation (mean proportion of days covered, 52%). Mean femoral neck T-score in the random subset of patients tested at 2 years favored the BHS arm (-0.55 vs -0.70, $P = .04$), although this did not meet the Bonferroni corrected $P < .025$ for this outcome. Fractures occurred in 29 patients (2.0%) of 1424 in the usual care group and 31 patients (1.8%) of 1688 in the BHS group ($P = .69$).

There were no falls or other safety concerns related to screening. One participant in the BHS arm had a drug interaction wherein the calcium supplementation interfered with levothyroxine absorption resulting in asymptomatic elevated thyroid-stimulating hormone levels. Prescriptions for proton pump inhibitors or histamine type 2 blockers were similar in BHS and usual care teams (562 [33.3%] vs 473 [33.2%], $P = .96$). No cases of osteonecrosis of the jaw or atypical femoral fractures were identified.

Discussion

With fracture rates rising worldwide, calls to identify cost-effective strategies to address the osteoporosis treatment crisis have become more urgent.³¹ This study demonstrated that a remote, centralized approach to case finding and osteoporosis management among men without a prior fracture but with 1 or more risk factors was readily adopted by patients and primary care clinicians and resulted in large improvements in

screening and treatment that are likely to translate to clinical benefit. It is notable that these high acceptance rates were achieved despite the study starting in the early days of the COVID-19 pandemic. This rigorous group-randomized trial confirms findings from a prior retrospective cohort study reporting that a centralized bone health team significantly increased the diagnosis of osteopenia and osteoporosis in older veterans³² and expands this by quantifying the impact on medication adherence and bone density outcomes.

Clinical guidelines vary on how to select men for primary osteoporosis screening. We targeted men with at least 1 comorbidity or high-risk medication associated with fractures. More than half of those screened by the BHS model were found to have osteopenia (349 [43.0%]) or osteoporosis (67 [8.1%]); these rates are comparable with women in the general US population for whom screening is routinely recommended (49% and 10%, respectively).³³ Currently, Medicare covers DXA scans for men for only limited conditions (hyperparathyroidism, glucocorticoids, androgen deprivation, osteopenia on radiography).³⁴ The program achieved high rates of medication persistence and adherence, both of which were associated with decreased fracture rates.^{35,36} Higher femoral neck T-scores at 2 years were observed in the BHS arm compared with the usual care arm; however, the difference did not meet the Bonferroni-corrected $P < .025$. This comparison was biased toward the null because 12% of patients in the BHS arm were ineligible for this outcome because less than 18 months had elapsed since their screening DXA. This occurred because of 9-month wait times for undergoing the initial DXA scan at one of the sites related to pandemic staffing issues. Total hip T-scores also favored patients in BHS group but the difference did not achieve statistical significance. Spine T-scores are difficult to interpret in the veteran population due to the high prevalence of osteophytes.³⁷

The BHS approach has both similarities and differences with the fracture liaison service model promoted as best

practice by the International Osteoporosis Foundation and the Bone Health and Osteoporosis Foundation. Both employ case finding of patients within a health system or region, management by a centralized team, and longitudinal follow-up. However, most fracture liaison services focus on patients who have already had a fracture, require in-person visits, and use more costly advanced practice providers as the prescribers.³⁸ In this study, the fully remote model used 1 full-time registered nurse coordinating care for nearly 2000 patients and used electronic consult support from an osteoporosis expert (approximately 10% effort). Thus, these findings demonstrate that a lower cost intervention targeting a wider group of at-risk individuals can achieve large and clinically important improvements in osteoporosis screening, treatment, and adherence. Our program had wide geographic reach; with more than half of the teams located in community-based clinics, the proportion screened from rural areas (33.7%) was similar to the overall proportion of rural veterans (33.0%). The VA is an integrated health care system with a strong commitment to preventive services and provides broader coverage of osteoporosis screening in men. However, with more than 60% of older adults enrolled in Medicare Advantage risk-based plans, many other health systems are implementing fracture liaison services and other population health management programs that this model can inform.³⁹

Limitations

This study has several limitations. First, although its strengths include the multicenter-group randomized design and blinding of outcomes assessment, the veteran population differs from nonveterans with higher fracture rates documented among both male and female veterans.^{40,41} Second, we used

comorbidities documented in the EHR to identify male veterans eligible for screening. This approach has been reported to have low sensitivity but good specificity, suggesting that a BHS program would likely miss individuals eligible for screening but have a relatively low rate of inappropriate screening.⁴² Third, we likely missed fracture outcomes because veterans sought care at non-VA facilities. Fourth, veterans were able to choose among VA and community DXA testing sites, precluding our ability to monitor quality control or use absolute bone mineral density as an outcome. Fifth, the clinical significance of the group difference in femoral neck T-score is unclear. Sixth, we were not able to adhere to our original analysis plan of employing a generalized linear mixed model because of the large number of usual care clinics with no screening events; the Poisson model allowed us to account for clustering for the T-score analysis, but not for the screening outcome. However, the between-group difference in screening rate was so large, accounting for clustering is unlikely to change our conclusions.

Conclusions

This cluster randomized clinical trial found that a centralized remote BHS is a feasible model and is associated with improving rates of osteoporosis screening, treatment, and adherence with high diagnostic yield. If found to be cost-effective, this care model could be readily spread as a low-touch approach to address the crisis in osteoporosis treatment in the US. This trial also supports the efficacy of risk-factor-based osteoporosis screening in men who have not yet sustained a fracture.

ARTICLE INFORMATION

Accepted for Publication: May 21, 2025.

Published Online: August 25, 2025.

doi:10.1001/jamainternmed.2025.4150

Author Contributions: Dr Colón-Emeric had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Concept and design: Colón-Emeric, Lee, Lyles, Zullig, Nelson, Adler.

Acquisition, analysis, or interpretation of data: Colón-Emeric, Lee, Lyles, Sloane, Pieper, Nelson, Adler.

Drafting of the manuscript: Colón-Emeric, Lee, Sloane, Pieper.

Critical review of the manuscript for important intellectual content: Lee, Lyles, Zullig, Sloane, Nelson, Adler.

Statistical analysis: Lee, Sloane, Pieper, Nelson.

Obtained funding: Colón-Emeric, Nelson.

Administrative, technical, or material support: Colón-Emeric, Lee, Zullig, Nelson, Adler.

Supervision: Lee, Nelson, Adler.

Conflict of Interest Disclosures: Dr Lyles reported being an equity owner in Faculty Connections LLC. Dr Zullig reported receiving consultancy fees from Eisai and Novartis outside the submitted work. Dr Adler reported receiving grants from the National Aeronautics and Space Administration and

serving as an associate editor of *Osteoporosis International*, and receiving honorarium from the University of Wisconsin outside the submitted work. No other disclosures were reported.

Funding/Support: This study was funded by grant 1I01HX002512-01 from the Veterans Affairs Health Systems Research.

Role of the Funder/Sponsor: The Veterans Affairs Health Systems had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Data Sharing Statement: See [Supplement 2](#).

Additional Contributions: We thank Ryan Crossgrove, MPH, and bone health service nurses Jyostna Jadhav, RN, BSN, and Ivuoma Igwe, RN, DNP, who contributed to data collection, study implementation, and regulatory adherence, all of whom were employed by the study.

REFERENCES

1. Bliuc D, Nguyen ND, Milch VE, Nguyen TV, Eisman JA, Center JR. Mortality risk associated with low-trauma osteoporotic fracture and subsequent fracture in men and women. *JAMA*. 2009;301(5):513-521. doi:10.1001/jama.2009.50

2. Ito K, Elkin EB, Girotra M, Morris MJ. Cost-effectiveness of fracture prevention in men who receive androgen deprivation therapy for localized prostate cancer. *Ann Intern Med*. 2010;152(10):621-629. doi:10.7326/0003-4819-152-10-201005180-00002
3. Müller D, Pulm J, Gandjour A. Cost-effectiveness of different strategies for selecting and treating individuals at increased risk of osteoporosis or osteopenia: a systematic review. *Value Health*. 2012;15(2):284-298. doi:10.1016/j.jval.2011.11.030
4. Schousboe JT, Schousboe JT. Cost effectiveness of screen-and-treat strategies for low bone mineral density: how do we screen, who do we screen and who do we treat? *Appl Health Econ Health Policy*. 2008;6(1):1-18. doi:10.2165/00148365-200806010-00001
5. Schousboe JT, Taylor BC, Fink HA, et al. Cost-effectiveness of bone densitometry followed by treatment of osteoporosis in older men. *JAMA*. 2007;298(6):629-637. doi:10.1001/jama.298.6.629
6. Colón-Emeric CS, Pieper CF, Van Houtven CH, et al. Limited osteoporosis screening effectiveness due to low treatment rates in a national sample of older men. *Mayo Clin Proc*. 2018;93(12):1749-1759. doi:10.1016/j.mayocp.2018.06.024
7. Alibhai SM, Yun L, Cheung AM, Paszat L. Screening for osteoporosis in men receiving

- androgen deprivation therapy. *JAMA*. 2012;307(3):255-256. doi:10.1001/jama.2011.2022
8. Cheng N, Green ME, Cheng N, Green ME. Osteoporosis screening for men: are family physicians following the guidelines? *Can Fam Physician*. 2008;54(8):1140-1141, 1141.e1-1141.e5.
 9. Nayak S, Greenspan SL. Cost-effectiveness of osteoporosis screening strategies for men. *J Bone Miner Res*. 2016;31(6):1189-1199. doi:10.1002/jbmr.2784
 10. Nicholson WK, Silverstein M, Wong JB, et al; US Preventive Services Task Force. Screening for osteoporosis to prevent fractures: US Preventive Services Task Force recommendation statement. *JAMA*. 2025;333(6):498-508. doi:10.1001/jama.2024.27154
 11. Watts NB, Adler RA, Bilezikian JP, et al; Endocrine Society. Osteoporosis in men: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab*. 2012;97(6):1802-1822. doi:10.1210/jc.2011-3045
 12. Liu H, Paige NM, Goldzweig CL, et al. Screening for osteoporosis in men: a systematic review for an American College of Physicians guideline. *Ann Intern Med*. 2008;148(9):685-701. doi:10.7326/0003-4819-148-9-200805060-00009
 13. Fuggle NR, Beaudart C, Bruyère O, et al. Evidence-based guideline for the management of osteoporosis in men. *Nat Rev Rheumatol*. 2024;20(4):241-251. doi:10.1038/s41584-024-01094-9
 14. Morin SN, Feldman S, Funnell L, et al; Osteoporosis Canada 2023 Guideline Update Group. Clinical practice guideline for management of osteoporosis and fracture prevention in Canada: 2023 update. *CMAJ*. 2023;195(39):E1333-E1348. doi:10.1503/cmaj.221647
 15. Adler R, Semla T, Cunningham F, Pogach L. The VHA male osteoporosis program: a national model for bone health. *Fed Pract*. 2012;29:31-37. <https://www.mdedge.com/fedprac/article/75623/geriatrics/vha-male-osteoporosis-program-national-model-bone-health>
 16. Lyles KW, Colón-Emeric CS, Magaziner JS, et al; HORIZON Recurrent Fracture Trial. Zoledronic acid and clinical fractures and mortality after hip fracture. *N Engl J Med*. 2007;357(18):1799-1809. doi:10.1056/NEJMoa074941
 17. Deardorff WJ, Cenzer I, Nguyen B, Lee SJ. Time to benefit of bisphosphonate therapy for the prevention of fractures among postmenopausal women with osteoporosis: a meta-analysis of randomized clinical trials. *JAMA Intern Med*. 2022;182(1):33-41. doi:10.1001/jamainternmed.2021.6745
 18. Hsu YH, Li CC, Liang FW, et al. Reduced all-cause mortality with bisphosphonates among post-fracture osteoporosis patients: a nationwide study and systematic review. *Clin Pharmacol Ther*. 2022;112(3):711-719. doi:10.1002/cpt.2645
 19. Elliot-Gibson V, Bogoch ER, Jamal SA, Beaton DE. Practice patterns in the diagnosis and treatment of osteoporosis after a fragility fracture: a systematic review. *Osteoporos Int*. 2004;15(10):767-778. doi:10.1007/s00198-004-1675-5
 20. Jaleel A, Saag KG, Danila MI. Improving drug adherence in osteoporosis: an update on more recent studies. *Ther Adv Musculoskelet Dis*. 2018;10(7):141-149. doi:10.1177/1759720X18785539
 21. Colón-Emeric CS, Lyles KW, House P, et al. Randomized trial to improve fracture prevention in nursing home residents. *Am J Med*. 2007;120(10):886-892. doi:10.1016/j.amjmed.2007.04.020
 22. Nayak S, Greenspan SL. How can we improve osteoporosis care? a systematic review and meta-analysis of the efficacy of quality improvement strategies for osteoporosis. *J Bone Miner Res*. 2018;33(9):1585-1594. doi:10.1002/jbmr.3437
 23. Barton DW, Piple AS, Smith CT, Moskal SA, Carmouche JJ. The clinical impact of fracture liaison services: a systematic review. *Geriatr Orthop Surg Rehabil*. 2021;12:2151459320979978. doi:10.1177/2151459320979978
 24. Li N, Hilgsmann M, Boonen A, et al. The impact of fracture liaison services on subsequent fractures and mortality: a systematic literature review and meta-analysis. *Osteoporos Int*. 2021;32(8):1517-1530. doi:10.1007/s00198-021-05911-9
 25. Colón-Emeric CS, Lee R, Pieper CF, et al. Protocol for the Models of Primary Osteoporosis Screening in men (MOPS) cluster randomized trial. *Contemp Clin Trials*. 2022;112:106634. doi:10.1016/j.cct.2021.106634
 26. Steuart Richards J, Lazzari AA, Teves Qualler DA, Desale S, Howard R, Kerr GS. Validation of the osteoporosis self-assessment tool in US male veterans. *J Clin Densitom*. 2014;17(1):32-37.
 27. Vrijens B, De Geest S, Hughes DA, et al; ABC Project Team. A new taxonomy for describing and defining adherence to medications. *Br J Clin Pharmacol*. 2012;73(5):691-705. doi:10.1111/j.1365-2125.2012.04167.x
 28. Prieto-Merino D, Mulick A, Armstrong C, et al. Estimating proportion of days covered (PDC) using real-world online medicine suppliers' datasets. *J Pharm Policy Pract*. 2021;14(1):113. doi:10.1186/s40545-021-00385-w
 29. Vashi AA, Urech T, Wu S, Tran LD. Community emergency care use by veterans in an era of expanding choice. *JAMA Netw Open*. 2024;7(3):e241626-e241626. doi:10.1001/jamanetworkopen.2024.1626
 30. Shepstone L, Lenaghan E, Cooper C, et al; SCOOP Study Team. Screening in the community to reduce fractures in older women (SCOOP): a randomised controlled trial. *Lancet*. 2018;391(10122):741-747. doi:10.1016/S0140-6736(17)32640-5
 31. Roux C, Briot K. The crisis of inadequate treatment in osteoporosis. *Lancet Rheumatol*. 2020;2(2):e110-e119. doi:10.1016/S2665-9913(19)30136-5
 32. Lawrence PT, Grotzke MP, Rosenblum Y, et al; The Bone Health Team. A team-based approach to improving osteoporosis care for primary care patients. *J Prim Care Community Health*. 2017;8(3):135-140. doi:10.1177/2150131916687888
 33. Looker AC, Melton LJ III, Harris TB, Borrud LG, Shepherd JA. Prevalence and trends in low femur bone density among older US adults: NHANES 2005-2006 compared with NHANES III. *J Bone Miner Res*. 2010;25(1):64-71. doi:10.1359/jbmr.090706
 34. Bone mass measurements. Center for Medicare & Medicaid Services. Accessed May 6, 2025. <https://www.medicare.gov/coverage/bone-mass-measurements>
 35. Landfeldt E, Ström O, Robbins S, Borgström F. Adherence to treatment of primary osteoporosis and its association to fractures—the Swedish Adherence Register Analysis (SARA). *Osteoporos Int*. 2012;23(2):433-443. doi:10.1007/s00198-011-1549-6
 36. Wade SW, Curtis JR, Yu J, et al. Medication adherence and fracture risk among patients on bisphosphonate therapy in a large United States health plan. *Bone*. 2012;50(4):870-875. doi:10.1016/j.bone.2011.12.021
 37. Knight SM, Ring EF, Bhalla AK. Bone mineral density and osteoarthritis. *Ann Rheum Dis*. 1992;51(9):1025-1026. doi:10.1136/ard.51.9.1025
 38. Napoli N, Ebeling PR, Kiel DP. Coordinating multidisciplinary care—improving outcomes after fragility fractures. *N Engl J Med*. 2025;392(2):109-111. doi:10.1056/NEJMp2409086
 39. Catlett K, Jarvis MS, Sullivan J, Skootsky SA, Dentzer S. Potential spillover effects on traditional Medicare when physicians bear Medicare Advantage risk. *Am J Manag Care*. Published online February 26, 2025. doi:10.37765/ajmc.2025.89686
 40. LaFleur J, Rillamas-Sun E, Colón-Emeric CS, et al. Fracture rates and bone density among postmenopausal veteran and non-veteran women from the Women's Health Initiative. *Gerontologist*. 2016;56(suppl 1)(suppl 1):S78-S90. doi:10.1093/geront/gnv677
 41. Hinojosa R, Hinojosa MS. Activity-limiting musculoskeletal conditions in US veterans compared to non-veterans: results from the 2013 National Health Interview Survey. *PLoS One*. 2016;11(12):e0167143. doi:10.1371/journal.pone.0167143
 42. Singh B, Singh A, Ahmed A, et al. Derivation and validation of automated electronic search strategies to extract Charlson comorbidities from electronic medical records. *Mayo Clin Proc*. 2012;87(9):817-824. doi:10.1016/j.mayocp.2012.04.015